BRAC UNIVERSITY
Department of Computer Science and Engineering

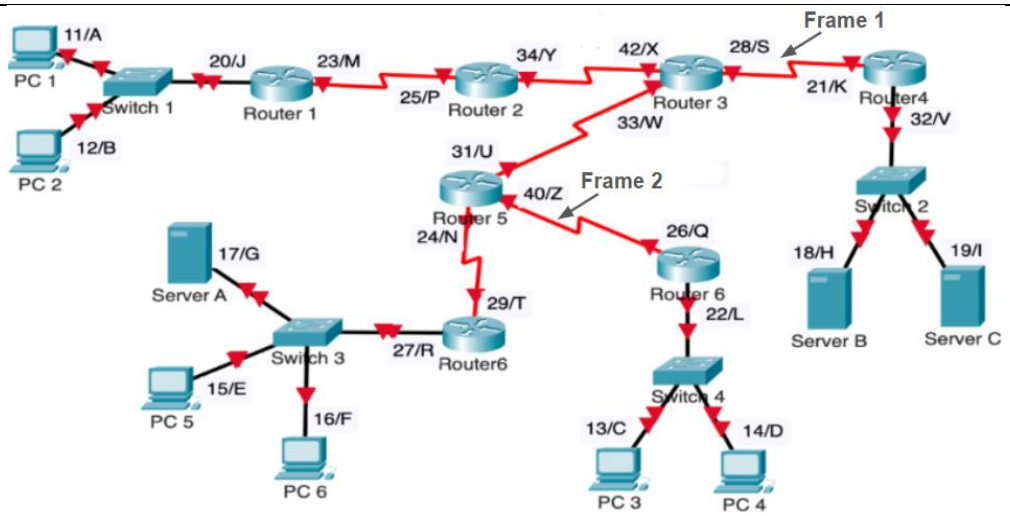
Examination: Semester Midterm | Semester: Spring 2026
Duration: 1 hour 20 min | Full Marks: 40

CSE 320/EEE361/ECE361: Data Communications

Answer the following questions.
Figures in the right margin indicate marks.

SET B

1. [CO1] a) [2+3]



Examine the network topology provided above. In this diagram, physical addresses are indicated by uppercase letters, and logical addresses are indicated by numbers. When assigning port numbers to end devices, use values from the dynamic port range (49152–65535).

i. **How many** LANs and WANs are there in total?

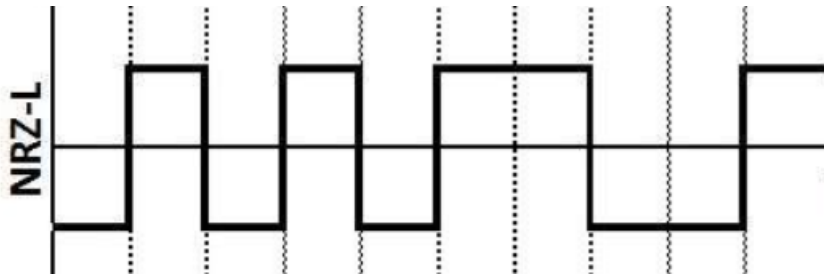
ii. A user at PC3 is requesting for a YouTube video hosted on Server C. The data is being travelling from PC3 to Server C (port 443). At a particular moment, the data is encapsulated in Frame 2. Considering this situation, **complete** Frame 2 below by writing the appropriate Destination and Source Physical, Logical and Port Addresses.

Frame 2	Destination MAC	Source MAC	Destination IP	Source IP	Destination Port	Source Port	Data	Trailer

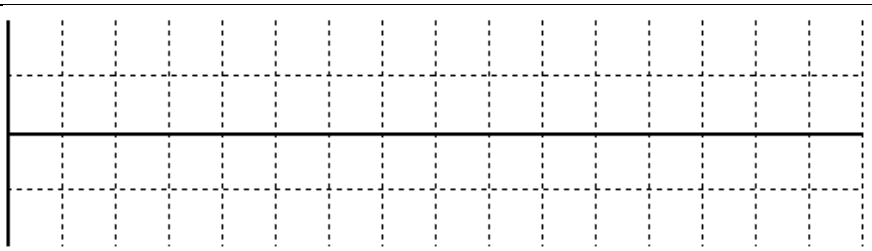
b) [3]

The Bangladesh Railway manages 5 stations in Dhaka city, connected using mesh topology. Each station has 3 departments (ticketing, control, maintenance), connected using star topology. Each department has 10 computers, connected using bus topology. **Draw** this hybrid topology and

		calculate the total number of links in the network. (Assume that each bus topology is considered as a single shared communication link, regardless of the number of connected devices.)	
	c)	Arif is downloading a document from a remote server. First, his device establishes a connection with the server using a three-way handshake. Once connected [1], the server sends data in segments with sequence numbers and port numbers. Each segment is encapsulated into packets containing source and destination addresses [2]. These packets are then converted into frames [3] and transmitted over the network as signals. At the receiving end, the process is reversed. The data is reassembled, checked for errors [4], and finally delivered to the application for display. **Identify** the TCP/IP layer (Layer name and number) responsible for each of the numbered steps [1] – [4].	[2]
2. [CO2]	a)	You are sending a periodic composite signal to your friend who lives 50 km apart through a transmission channel. The signal can be decomposed into three sine waves with time periods of 400 μs, 200 μs and 60 μs, respectively. The sine wave with the highest time period has a peak amplitude of 10V, the one with the lowest time period has a peak amplitude of 5V and the remaining one has a peak amplitude of 7V. **Draw** the composite analog signal in the frequency-domain representation.	[2]
	b)	 Suppose, a signal is being transmitted along a transmission line from point A to point F. The power of the signal at point A is 60 mW. The distance between point A and point B is 10 Km and the power loss rate from point A to point B is 0.3 mW/Km. To prevent excessive signal degradation, an amplifier is placed between point B and point C and the power gets amplified 3 times here. The attenuation coefficient from point C to point D is 1.5 dB and the power loss rate is 0.75 dB/Km. Again, an amplifier is placed between point D and point E and the power gets amplified 4 times here. The distance from point E to point F is 4 Km and the loss along this path is 0.3 dB/Km. i. **Calculate** the distance between point C to point D. ii. **Calculate** the power of the signal at point F. **Determine** whether the signal was amplified or attenuated from point A to point F.	[3+2+1]
	c)	Suppose, a document file containing 7 pages is being transmitted over a noisy channel at a maximum bit rate of 150 kbps. Each page contains 65 characters, and the file is encoded using ASCII (8 bits per character). The noise power is 60 times weaker than the signal power.	[2+2+1]

		i. **Calculate** the bandwidth of the channel. ii. If the actual bit rate of the channel is 25% less than the theoretical maximum capacity, **determine** the number of signal levels. iii. **Compute** the transmission delay for sending the entire document file over this channel.	
	d)	Doubling the SNR does not double the bit rate. Agree or disagree? **Justify** your answer with proper reasons.	[2]
3. [CO2]	a)	Baseline wandering is a common problem in both NRZ-I and NRZ-L; however, it is twice in NRZ-L scheme. **Explain** why? Hint: Inspect the nature of the encoding scheme and what happens for 0s and 1s.	[2]
	b)	Suppose you are applying 3B/6B block coding in your network system. To get certain advantages, you are following some rules to map the 3-bit combinations with the 6-bit combinations. **Observe** the below mapping table and **mention** 3 **rules** that are applied to make a 6-bit codeword in the given table.	[6]
	c)	 i. **Decode** the signal above and **generate** the binary bit stream using **NRZ-L** line coding scheme. ii. Now from the extracted binary bits, **draw** the signal in the given graph using an appropriate line coding scheme so that there is no synchronization problem. (draw it in the question paper only)	[2+2]

Dataword	Codeword
000	101001
001	010110
010	100110
011	011001
100	110010
101	101010
110	010101
111	011010

			
	d)	**Apply** a technique that can resolve both the synchronization issues and the DC problem for a long sequence of 0's without requiring any extra bandwidth. **What** is the name of the technique? **Encode** the following binary bits **using the graph below**: **01000000111** 	[3]

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